

Report – SemEval 2026 Task 5

Rating Plausibility of Word Senses in Ambiguous Stories through Narrative Understanding

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Abstract

While large language models excel at semantic reasoning, their discrete token-based outputs introduce limitations for fine-grained regression tasks requiring continuous scoring. We address graded word-sense plausibility estimation by reformulating it as a Natural Language Inference regression problem, adapting DeBERTa-v3-large with NLI pretraining and a regression head to predict continuous plausibility scores from story-sense pairs. Evaluated against BERT, vanilla DeBERTa, SmolLM variants and state-of-the-art LLMs under various prompting strategies, the NLI-finetuned model achieves superior rank correlation and alignment with human judgments. While several baselines collapse toward mean predictions and LLMs show unstable prompting sensitivity, our findings establish NLI-informed pretraining as highly effective for narrative plausibility regression, highlighting fundamental LLM limitations for word sense disambiguation.

Code: github.com/NiccoloAntonelliDziri/LLM-SemEval-T5

1 Introduction

Word sense disambiguation has long been a fundamental challenge in natural language processing, requiring systems to determine which meaning of a polysemous word is intended in a given context. Traditional approaches typically assume that only one sense is correct for any given usage, framing the task as a classification problem (e.g. (Navigli, 2009)). However, this binary view doesn't fully capture how humans perceive ambiguity in language, where multiple interpretations may be simultaneously plausible to varying degrees.

SemEval 2026 Task 5 addresses this challenge by focusing on rating the plausibility of different word senses within ambiguous sentences through narrative understanding. Rather than simply selecting the correct interpretation, this task requires systems to evaluate how plausible each potential sense is within the context of short narrative stories. This approach acknowledges that ambiguity, underspecification, and individual linguistic experience all influence how humans interpret meaning, with narrative coherence playing a crucial role in these judgments.

In this report, we compare specialized models, such as DeBERTa-v3 (He et al., 2021), which are trained for specific tasks, with larger, state-of-the-art LLMs like Llama 3.1 and Deepseek-r1. Our focus is on two key aspects: how well these models rank information similarly to humans (spearman correlation) and how consistent their results are with human expectations (accuracy within standard deviation).

We demonstrate that while LLMs possess vast linguistic knowledge, they often fail to capture the human intuition, a task where specialized regression-tuned models remain superior.

2 Task description

The task is built upon the AmbiStory dataset (Gehring and Roth, 2025), designed to probe how narrative progression resolve or maintain lexical ambiguity. Each instance consists of a five-sentence narrative featuring a target homonym.

Each story is structured into three parts:

- **Precontext:** three sentences that establish the setting and relevant events.
- **Ambiguous sentence:** the fourth sentence

containing a homonym with two widely different candidate senses.

- **Ending:** an optional fifth sentence, where one of two possible endings (or no ending) may be provided and often biases the interpretation toward one sense.

For each story setup, participants rate the plausibility of each candidate sense on a scale from one to five and at least five ratings were collected per story instance.

Given the story context and a candidate sense definition, our system must output a plausibility score in [1, 5] for that sense. Evaluation uses two complementary metrics:

- **Spearman correlation:** correlation between the model predictions and the mean human plausibility scores;
- **Accuracy within standard deviation:** proportion of predictions that fall within one standard deviation of the mean human rating, reflecting agreement under varying annotator consensus.

3 Methodology

3.1 The Gold Standard

The core idea of our approach was to adapt the task into a Natural Language Inference (NLI) problem, using an hypothesis-premise pair:

- **Premise:** The full story context, composed of precontext, target sentence, and ending (if available).
- **Hypothesis:** A structured sense definition: *The definition of ‘{homonym}’ is: ‘{judged_meaning}’ as in the following sentence: ‘{example_sentence}’.*

We use a pre-trained DeBERTa-v3-large model(He et al., 2021), specifically the variant fine-tuned(Laurer et al., 2024) on multiple NLI datasets. We adapted this model for our regression task by replacing the original three-class classification head with a single-output regression head.

This formulation allows the model to leverage its pre-trained understanding of semantic entailment to assess the plausibility of each word sense within the given narrative context.

3.2 Alternative Baseline Architectures

To quantify the gain provided by DeBERTa-v3’s architecture, we benchmarked several alternative families against our primary NLI-finetuned DeBERTa model:

BERT-base (Devlin et al., 2018) We utilized this state-of-the-art model as a baseline for the encoder family. This comparison serves to measure the impact of parameter scale (110M vs 435M) and the performance ceiling of early transformer architectures on narratively dense tasks.

Vanilla DeBERTa-v3-large To specifically quantify the gain provided by the NLI-specific pre-finetuning, we trained a "vanilla" version of the DeBERTa-v3-large model. This version shares the advanced attention mechanism of our primary model but lacks the internal representations of semantic entailment provided by the NLI datasets, allowing us to evaluate the gain of the transfer learning.

SmolLM (Allal et al., 2024) Shifting from encoder-based masked models to decoder-only (causal) architectures, we finetuned the three variants of the SmolLM series. Specifically:

- **135M:** A minimal-parameter baseline for efficient edge deployment.
- **360M:** A mid-range model comparable in size to DeBERTa-v3-large but with a causal objective.
- **1.7B:** A model that explores the benefits of significantly larger parameter counts in a "small" language model format.

3.3 Large Language Model (LLM) Benchmarking

Finally, we evaluated a broad range of state-of-the-art LLMs to assess their zero-shot and few-shot performance on the task, considering both instruction-tuned and reasoning ("thinking") variants.

Three primary prompting tiers were implemented:

- **Zero-Shot Standard:** Models were prompted as "Expert NLU Annotators" and required to output a JSON object containing the 1-5 score.
- **Few-Shot Prompting:** Models received five in-context examples constructed from the training dataset (one for each score) to prime the task.

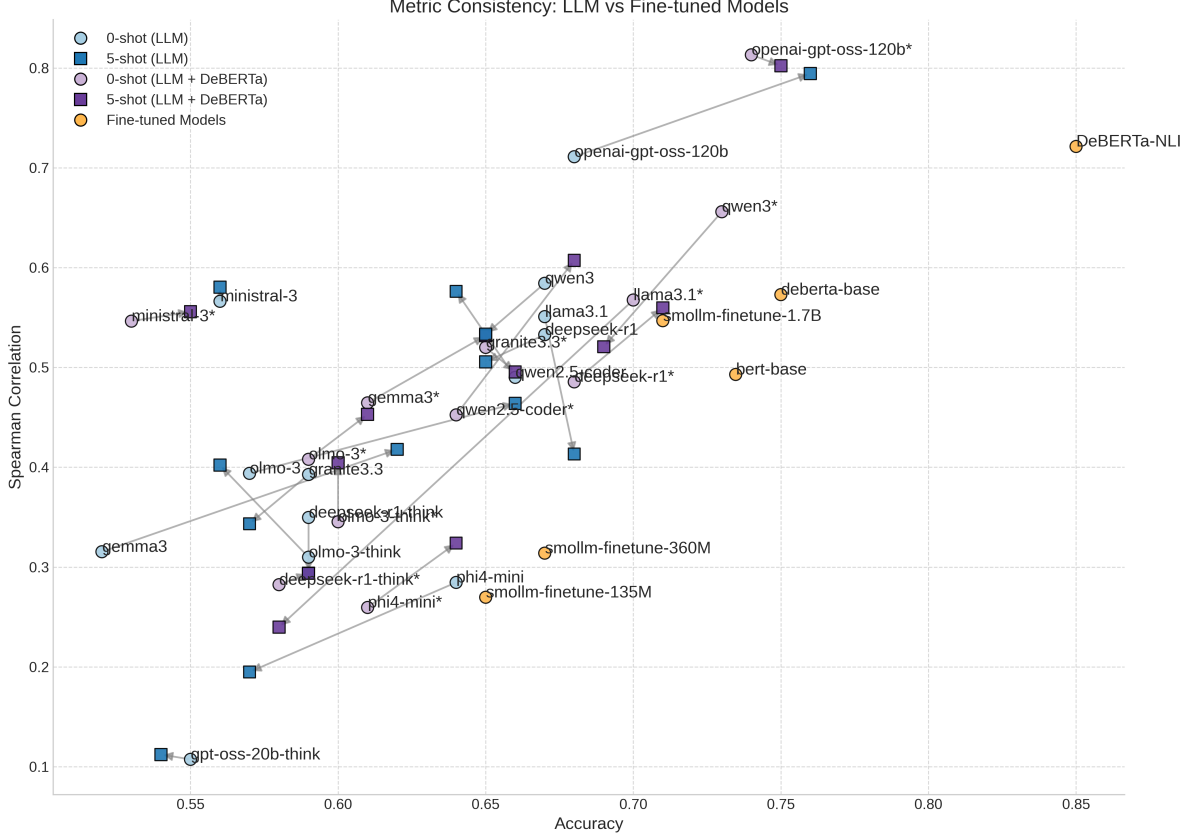


Figure 1: Metric consistency: accuracy and spearman on a randomly selected 100 elements. The '*' next to model names indicates that they were run with DeBERTa injection (purple dots and squares).

- **Hybrid Contextualization:** We injected the raw DeBERTa-v3-NLI prediction into the LLM prompt as an "Additional Context" signal. The goal was to test if an LLM could act as a refiner, using its narrative understanding to adjust the initial score provided by the specialized encoder.

For reasoning models with significantly higher inference costs, we evaluated on a fixed set of 100 randomly selected instances to make the results comparable.

To ensure results were comparable across architectures, all LLMs were set to temperature=0 for deterministic scoring.

4 Experiments

The AmbiStory dataset provides two files: test.json and dev.json. To be consistent with the first phase of the SemEval task, and to prevent data leakage, we used test.json to build our training and validation sets (90/10), and we reserved dev.json for testing.

Models were trained and LLMs were evaluated

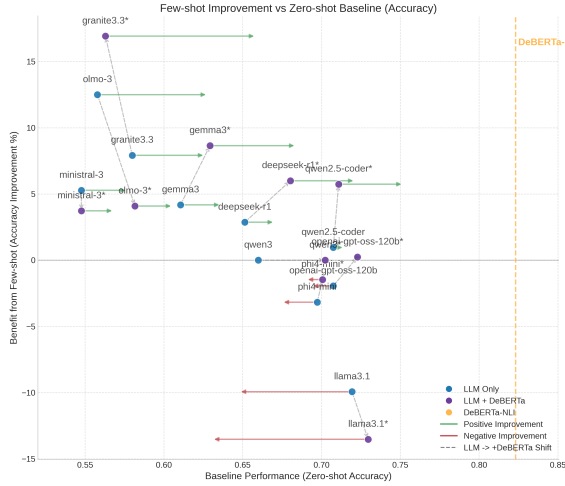
on an NVIDIA GeForce RTX 4060 laptop GPU, except for gpt-oss-120b, whose inference was run via Groq (free plan).

5 The Metric Consistency Landscape

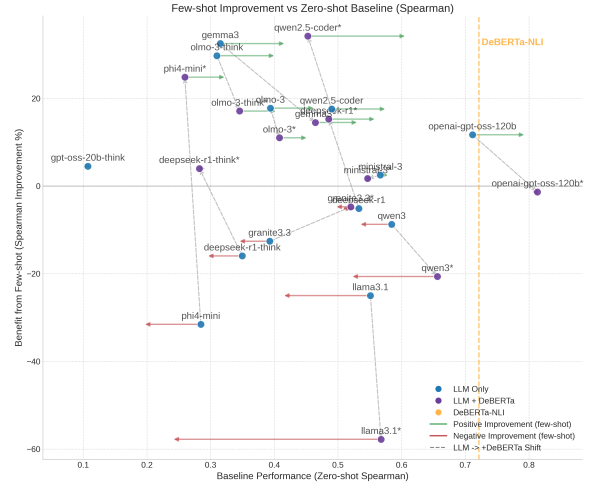
The core of our evaluation lies in the relationship between spearman correlation and accuracy within standard deviation. Figure 1 visualizes this landscape, mapping each model's ability to rank plausibility (y-axis) against its ability to remain within human consensus boundaries (x-axis).

5.1 Finetuned models

A first observation from Figure 1 is the grouping of finetuned models (BERT family and SmolLM series indicated as orange dots) along a line in the high-accuracy/low-correlation region. When these models are finetuned on the AmbiStory regression task with an MSE objective, they can improve their loss by gravitating toward a conservative, near-constant prediction close to the global mean (around 3). This can yield a relatively high accuracy within standard deviation (many predictions fall near the typical rating range) while still



Accuracy within Std. Dev.



Spearman correlation

Figure 2: Few-shot learning potential on a randomly selected subset of 100 examples. We report accuracy within standard deviation (left) and spearman correlation (right) across the four prompting configurations. The arrows point towards the values of the five-shot prompting. The grey dotted lines link the models with and without DeBERTa.

failing to capture the monotonic ranking of senses, leading to low spearman correlation.

For the BERT family, gains appear to be driven by architecture choices (DeBERTa vs. BERT) and, crucially, by the intermediate fine-tuning regime (NLI-supervised weights vs. vanilla weights). DeBERTa-NLI (top-right) suggests that encoder-based models can achieve both high accuracy and strong rank correlation when they benefit from additional training on an entailment-style objective, which is closely related to our premise-hypothesis formulation.

The finetuned SmolLM variants reveal a consistent scaling trend within decoder-only architectures: the 1.7B variant substantially outperforming both 360M and 135M baselines. However, even the largest SmolLM variant falls short of DeBERTa-NLI, suggesting that causal language modeling, while capable of learning regression, has fundamental difficulties with fine-grained plausibility estimation. This architectural limitation foreshadows the broader struggle we observe across instruction-tuned LLMs: if even optimized decoder models trained explicitly on the task cannot match entailment-informed encoders, then general-purpose language models are unlikely to excel at this regression problem.

5.2 Not all LLMs are equal

To analyze the LLMs, we benchmarked each one in four distinct configurations to observe their move-

ment across the landscape: zero-Shot, five-Shot, zero-shot + DeBERTa injection, and five-Shot + DeBERTa injection in order to try to bridge the gap between the specialized precision of DeBERTa and the broad reasoning of LLMs.

Contrary to the expectation of a universal "horizontal shift" toward higher accuracy upon providing DeBERTa context, we observed in Figure 2 a stratified response. Some models like Gemma3 or Granite3.3 showed significant alignment gains when injected with DeBERTa scores, essentially adopting the encoder’s scale as a prior. However, other models, such as Llama3.1 or Olmo3, exhibited signal interference: the injection of the external score occasionally caused the model to vacillate between its internal semantic logic and the teacher signal, leading to a decrease in both accuracy and correlation. This suggests that for some models, the DeBERTa prediction acts as an adversarial noise rather than an helpful support.

We observed a similar effect with few-shot prompting in some cases (Phi4-mini or Qwen3), where adding examples degraded performance instead of improving it.

While few-shot prompting is traditionally expected to improve task alignment, our experiments revealed a performance degradation in several LLMs when transitioning from zero-shot to five-shot configurations. We hypothesize that this degradation is rooted in an attenuation of lexical focus. In a zero-shot setting, the model’s attention is

strictly concentrated on the single target homonym and its relationship to the provided candidate sense within one story. However, a five-shot prompt for AmbiStory contains roughly 25 to 30 sentences of extraneous narrative context.

Overall, DeBERTa score injection is most reliable in the zero-shot setting or when paired with explicit reasoning prompts. In contrast, for instruction-tuned LLMs, combining few-shot exemplars with a numeric teacher prior can introduce conflicting signals, often reducing both accuracy and rank correlation.

5.3 Gpt-oss-120b performance

Notably, the large open-source model ‘openai-gpt-oss-120b’ shows a pattern where its ability to rank candidate senses can be competitive with the DeBERTa-NLI once given additional context, while its absolute accuracy within the human standard-deviation range is slightly lower. For instance, in our 100-instance evaluation ‘openai-gpt-oss-120b’ (zero-shot) reports spearman around 0.685 and accuracy around 0.707, whereas DeBERTa-NLI yields spearman around 0.693 and Accuracy around 0.823. When the DeBERTa prediction is injected into the LLM prompt, spearman increases to around 0.737 while accuracy remains comparatively unchanged, hinting that large models can improve ranking but not necessarily calibration.

These results suggest that very large LLMs can come close on ranking under carefully tuned prompting, but that their behaviour remains sensitive to prompting choices and auxiliary signals; in contrast, the small encoder-only DeBERTa-NLI remains the strongest and most stable model overall. Crucially, it achieves this with orders of magnitude fewer parameters, making it substantially more efficient in terms of performance per model size.

5.4 Impact of the ending sentence

To evaluate the effect of the ending sentence, we computed metrics separately for examples that include an ending and those that do not. The DeBERTa-NLI model shows nearly identical accuracy (82.65% without vs 82.14% with), but a clear improvement in correlation (spearman 0.651 vs 0.707), indicating that the ending helps the model evaluate candidate choices more like human judgments even when it rarely changes the single best prediction. This pattern suggests that the ending provides additional disambiguating cues, which

helps the model better distinguish between the candidate senses.

6 Discussion and comments

While DeBERTa regression head provides a continuous score, LLMs are fundamentally optimized for token selection and discrete labeling. When prompted to use the output of DeBERTa-NLI, the LLMs act as discretizers. This process destroys the granularity of the regression.

We also report a random baseline that samples plausibility scores according to the empirical label distribution observed in the training set (i.e. not from a uniform distribution). Under this distribution-matched guessing strategy, performance behaves as expected, with near-zero spearman correlation and an accuracy of around 46%. We additionally consider a degenerate baseline that always predicts the central score 3; in this case, spearman correlation is undefined (NaN) due to the lack of variance in the predictions, while accuracy rises to 53%. Taken together, these baselines confirm that the task is non-trivial (even a prior-matched or mean-only strategy remains weak) and that every model learns or understands meaningful patterns beyond label-frequency effects.

7 Conclusion

We reformulate graded sense-plausibility estimation as an NLI-style regression task and show that an encoder-only DeBERTa-v3 model with NLI pre-training achieves the best overall performance, outperforming both encoder baselines (BERT, vanilla DeBERTa) and decoder-only SmolLM variants. Across a broad LLM benchmark, results are comparatively unstable and sensitive to prompting and score-injection strategies, indicating that general-purpose LLMs remain less reliable for fine-grained continuous scoring. Overall, the strong performance of DeBERTa-NLI at a much smaller parameter budget highlights the effectiveness and efficiency of entailment-informed encoders for narrative plausibility regression. More broadly, the combination of narrative reasoning and continuous plausibility scoring makes this task a promising candidate for future LLM benchmarking.

References

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A Appendix

This appendix provides additional details and qualitative examples omitted from the main paper for space.

A.1 Zero-Shot Prompt

System prompt:

You are an expert NLU annotator. Your job is to
→ rate how plausible a candidate meaning
→ (sense)
is for the \textit{HOMONYM} used in the target
→ sentence within the short story.

Return ONLY a single JSON object with one key:
→ "score" and an integer value 1, 2, 3, 4 or 5.
Integer mapping:
1 = Definitely not
2 = Probably not
3 = Ambiguous / Unsure
4 = Probably yes
5 = Definitely yes

User prompt:

[STORY]
{full_story_text}

[HOMONYM]
{homonym}

[CANDIDATE SENSE]
{sense_text}

[TASK]
Based on the STORY above, decide how plausible it
→ is that the HOMONYM is used with the
CANDIDATE SENSE in the target sentence.

USER PROMPT:
[STORY]
{full_story_text}

[HOMONYM]
{homonym}

[CANDIDATE SENSE]
{sense_text}

[TASK]
Based on the STORY above, decide how plausible it
→ is that the HOMONYM is used with the
CANDIDATE SENSE in the target sentence.

Return ONLY a single JSON object with one key
→ "score" and an integer value (1-5)
as described by the system message. Example
→ output: {"score": 3}}

A.2 Few-Shot Prompt

System prompt:

You are an expert NLU annotator. Your job is to
→ rate how plausible a candidate meaning
→ (sense)
is for the HOMONYM used in the target sentence
→ within the short story.

Return ONLY a single JSON object with one key:
→ "score" and an integer value 1, 2, 3, 4 or 5.
Integer mapping:
1 = Definitely not
2 = Probably not
3 = Ambiguous / Unsure
4 = Probably yes
5 = Definitely yes

The response must be a JSON object and nothing
→ else, for example: {"score": 4}

[EXAMPLES]
{few_shot_examples}

User prompt:

Now, label this new instance:

[STORY]
{full_story_text}

[HOMONYM]
{homonym}

[CANDIDATE SENSE]
{sense_text}

[TASK]
Based on the STORY above, decide how plausible it
→ is that the HOMONYM is used with the
CANDIDATE SENSE in the target sentence.

Return ONLY a single JSON object with one key
→ "score" and an integer value (1-5)
as described by the system message. Example
→ output: {"score": 3}

A.3 Prompts with DeBERTa Injection

In the DeBERTa-injection setting, the system prompts remain unchanged. We modify only the user prompt by adding the following text:

[ADDITIONAL CONTEXT]
A DeBERTa model (Accuracy:
→ {deberta_accuracy:.2f}, Spearman Correlation:
→ {deberta_spearman:.2f}) predicted a score of
→ {deberta_prediction:.2f} for this example.
You can use this information to guide your
→ decision, but rely on your own judgment if
→ the context strongly suggests otherwise.